INFO8006 Introduction to Artificial Intelligence

Exam of January 2025

Instructions

- The exam lasts for 4 hours.
- You are allowed to use a calculator during the exam, but documents of any kind are forbidden.
- The last two pages can be used for scratch work or for extra space. If you want work done there to be graded, mention where to look **in big letters with a box around them**, on the page with the question.
- Write your last name, first name, and ULiège ID on the first page. Write only your ULiège ID on all the other pages.
- Before handing in your exam, **sort all the pages according to the page numbers** (even if you used additional pages to answer a question).

Good luck!

LAST NAME, FIRST NAME (in uppercase): ____________________

ULiège ID (s201234): ____________________

Question 1 [4 points] Multiple choice questions. Choose one of the four choices by filling in its circle. Correct answers are graded $+\frac{4}{10}$, wrong answers are graded $-\frac{2}{15}$ and the absence of answers is graded 0. The total of your grade for Question 1 is bounded below at 0/4.

1. An agent is rational if ...
   - ○ its decision process mimicks the human brain.
   - ○ its behavior is human-like.
   - ○ logical reasoning is used to make decisions.
   - ○ its decisions are made to maximize its expected performance.
2. An offline problem-solving agent ...
   - ○ recomputes its plan at each timestep.
   - ○ does not have access to a transition model.
   - ○ solves search problems in partially observable environments.
   - ○ computes a plan once and then executes it eyes closed.
3. The EXPECTIMINIMAX algorithm ...
   - ○ is used for training convolutional neural networks.
   - ○ is a variant of the MINIMAX algorithm for stochastic games.
   - ○ randomizes the execution of the MINIMAX algorithm to make it faster.
   - ○ is optimal, in expectation, if the heuristic function is admissible.
4. Which is false? The Kolmogorov's axioms state or imply that ...
   - ○ the probability $P(\omega) \geq 0$ for all sample points $\omega \in \Omega$.
   - ○ the probability $P(\omega) \leq 1$ for all sample points $\omega \in \Omega$.
   - ○ the probability $P(\Omega) = 1$.
   - ○ the probability $P(\{\omega_1, \ldots, \omega_n\}) = \prod_{i=1}^n P(\omega_i)$ for any set of mutually exclusive sample points $\{\omega_1, \ldots, \omega_n\} \subseteq \Omega$.
5. Let X, Y and Z be random variables such that $P(x, y, z) = P(x)P(y)P(z|x, y)$. Which is false?
   - ○ The random variables X, Y and Z form a v-structure.
   - ○ The random variables X and Y are conditionally independent given Z.
   - ○ The random variables X and Y are independent.
   - ○ $P(x, y, z) = P(z)P(x|z)P(y|x, z)$.
6. For a set $\mathbf{d}$ of observations and a probabilistic model $P(\mathbf{d}|\theta)$, the maximum likelihood estimate of the parameter θ is ...
   - ○ $\theta^* = \arg\max_\theta P(\theta)$.
   - ○ $\theta^* = \arg\max_\theta P(\mathbf{d}|\theta)$.
   - ○ $\theta^* = \arg\max_\theta P(\theta|\mathbf{d})$.
   - ○ $\theta^* = \arg\max_\theta P(\mathbf{d})$.
7. Let $w_{1:t-1}$ be a sequence of $t-1$ words forming the beginning of a sentence. Learning to predict the next word w_t is an example of ...
   - ○ classification problem.
   - ○ regression problem.
   - ○ segmentation problem.
   - ○ adversarial search problem.

8. In a deep neural network with `ReLU` activation functions in its hidden layers and no activation function in its output layer, ...
   - ◯ the input must be positive for the output to be positive.
   - ◯ the output is always positive, regardless of the input.
   - ◯ the function represented by the network is piecewise linear.
   - ◯ chaining multiple layers does not increase the representational power.
9. A Markov Decision Process is defined as a ...
   - ◯ tuple (S, A, P, Q) where S is the set of states, A the set of actions, P the transition model and Q the Q-function.
   - ◯ tuple (S, A, P, R) where S is the set of states, A the set of actions, P the transition model and R the reward function.
   - ◯ tuple (S, A, P, V) where S is the set of states, A the set of actions, P the transition model and V the value function.
   - ◯ tuple (S, A, P, π) where S is the set of states, A the set of actions, P the transition model and π the policy.
10. The Bellman optimality equations generalize to state-action-value $Q(s, a)$ as ...
    - ◯ $Q(s,a) = R(s) + \gamma \sum_{s'} P(s'|s,a) Q(s', a)$.
    - ◯ $Q(s,a) = R(s) + \gamma \sum_{s'} P(s'|s,a) Q(s', \pi(s'))$.
    - ◯ $Q(s,a) = R(s) + \gamma \sum_{s'} P(s'|s,a) \max_{s'} Q(s', a')$.
    - ◯ $Q(s,a) = R(s) + \gamma \sum_{s'} P(s'|s,a) \max_{a'} Q(s', a')$.

Question 2 [4 points] Pacman is back and needs your help to chase the ghosts! In this version of the game, Pacman navigates an $M \times N$ maze and can move in the four cardinal directions {`north`, `south`, `east`, `west`} unless there is a wall in the way. Ghosts are hidden in the maze and become revealed once Pacman is in line of sight. That is, a ghost is revealed if it is in the same row or column as Pacman with no wall in between. Pacman can eliminate a ghost by stepping onto its square and the game ends when all ghosts are eliminated. The goal is to find and eliminate all G ghosts as quickly as possible.

Notes:

- Ghosts are not moving and their position is known for the entire duration of the game once they have been revealed.
- Although Pacman cannot see through walls, he can see through ghosts and reveal all ghosts in the same row or column as the one he is currently on (up to the walls).

(a) Define a minimal state space for solving this search problem and determine its size.

(b) Given the state space defined above, define an initial state, an action function, a transition model, a goal test, and a step cost function for this search problem. You can assume that Pacman always starts at the top-left corner of the maze.

(c) For each heuristic below, determine whether it is admissible or not. Justify your answer.

- $h_1(s)$: the shortest Manhattan distance from Pacman's current position to any revealed ghost.
- $h_2(s)$: the sum of the Manhattan distances from Pacman's current position to every revealed ghost.
- $h_3(s)$: the number of ghosts still on the board.
- $h_4(s)$: the minimum number of moves to reveal a row or a column (possibly) containing a ghost.

(d) The partial observability of the environment creates a dilemma for Pacman, who must decide between exploring the maze to reveal the ghosts or eliminating the ghosts already revealed. Propose a strategy to address this dilemma and discuss its strengths and weaknesses.

Question 3 [4 points] The "Foire de Liège" has a new attraction called "Super Spring Ultra Pro Max XXL" which consists in a ball attached to a spring on a platform. The participants take seat in the ball locked at some position. The ball is then released and pulled by the spring which makes it oscillate back and forth. After a few seconds, the ball is stopped by magnetic brakes. You notice that the final position of the ball is different each time. As the ball is opaque, you wonder if it is possible for the participants to guess where the ball stopped, given their perception of acceleration.

You more or less remember your Newtonian mechanics class and model the movement of the ball as a series of transitions

$$\begin{aligned} p_t &= p_{t-1} + \Delta t \, \dot{p}_{t-1} + \frac{1}{2}\Delta t^2 \, \ddot{p}_{t-1}, \\ \dot{p}_t &= \dot{p}_{t-1} + \Delta t \, \ddot{p}_{t-1}, \\ \ddot{p}_t &= -\kappa \, p_{t-1} - \eta \, \dot{p}_{t-1}, \end{aligned}$$

where p_t, $\dot{p}_t$ and $\ddot{p}_t$ are respectively the position, velocity and acceleration of the ball at timestep t, Δt is the time elapsed between $t-1$ and t, κ is the stiffness of the spring, and η is the friction coefficient of the platform. You estimate that the ball starts $10.0 \pm 0.5\,\mathrm{m}$ at the left of the spring with a negligible speed $0.0 \pm 0.1\,\mathrm{m\,s^{-1}}$ and acceleration $0.0 \pm 0.1\,\mathrm{m\,s^{-2}}$. Finally, you assume that the human perception of acceleration follows an unbiased Gaussian distribution.

(a) You wish to predict the state of the ball given the the perceptions of a participant. Define and instantiate the components of a Kalman filter for this problem.

(b) Express the distribution $p(x_t|e_{1:t})$ with respect to the components defined previously. Use the Gaussian identities in Appendix A if needed.

(c) Represent the transition and sensor models as a dynamic Bayesian network.

Question 4 [4 points] We consider a neural network with one hidden layer, taking as input a scalar $x \in \mathbb{R}$ and producing as output a scalar $y = \mathrm{NN}_1(x;\theta) \in \mathbb{R}$. The neural network is defined as

$$\begin{aligned} h_0 &= \mathrm{ReLU}(w_0 x + b_0), \\ h_1 &= \mathrm{ReLU}(w_1 x + b_1), \\ y &= v_0 h_0 + v_1 h_1 + c, \end{aligned}$$

where $\theta = (w_0, b_0, w_1, b_1, v_0, v_1, c)$ is the set of parameters and $\mathrm{ReLU}(x) = \max(x, 0)$ is the rectified linear unit activation function.

(a) For $\theta = (2, 0, 1, -0.5, 1, -4, 0)$, draw the function $y = \mathrm{NN}_1(x;\theta)$ for $x \in [0, 1]$.

(b) This neural network has limited capacity. For this reason, we decide to feed the output of the network to a second neural network NN_2 with the same architecture as NN_1.

Draw the function $y = \mathrm{NN}_2(\mathrm{NN}_1(x;\theta_1);\theta_2)$ for $x \in [0, 1]$ using the parameters $\theta_1 = (2, 0, 1, -0.5, 1, -4, 0)$ and $\theta_2 = (2, 0, 1, -0.25, -1, \frac{10}{3}, -0.5)$.

(c) Instead of chaining NN_1 and NN_2, we decide to use a single neural network NN_3 with two hidden layers, each with two neurons.

(i) Derive the forward pass equations (from input to output, with each intermediate value) of this new neural network.

(ii) Motivate whether this new architecture has more or less capacity than the composition of NN_1 and NN_2.

(d) We now decide to adapt the architecture of NN_3 to solve a binary classification problem.

(i) Explain how you should change the architecture to produce appropriate outputs for this new task.

(ii) Following the principles of maximum likelihood estimation, derive the loss function to train the neural network on a dataset of N samples $\{x_i, y_i\}_{i=1}^N$.

Question 5 [4 points] In Micro-Blackjack, you repeatedly draw a card (with replacement) that is equally likely to be a 2, 3, or 4. At each step, if the total score of the cards you have drawn is less than 6, you can either draw (d) or cash (c). Otherwise, you can only cash. When you cash, the game stops and you receive a reward equal to the total score of the cards you have drawn (up to 5) plus 1, or zero if you get a total of 6 or higher. When you draw, you receive a reward of zero. The discount factor is $\gamma = 1$.

(a) Formalize Micro-Blackjack as an MDP.

(b) Describe the Value iteration algorithm, with pseudo-code.

(c) Derive the optimal policy for this MDP using Value iteration.

(d) If you were not told the rules of the game, what algorithm would you use to obtain an optimal policy?

Appendix A. Gaussian identities (Särkkä, 2013).

(a) If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C}\\ \mathbf{C}^T & \mathbf{B}\end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned}
p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{a}, \mathbf{A})\\
p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{b}, \mathbf{B})\\
p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x}|\mathbf{a} + \mathbf{C}\mathbf{B}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C}\mathbf{B}^{-1}\mathbf{C}^T)\\
p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}).
\end{aligned}$$

(b) If the random variables $\mathbf{x}$ and $\mathbf{y}$ have Gaussian probability distributions

$$\begin{aligned}
p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{P})\\
p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}),
\end{aligned}$$

then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{m}\\ \mathbf{H}\mathbf{m} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\mathbf{P} & \mathbf{P}\mathbf{H}^T\\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Extra page 1 / 2.

Extra page 2 / 2.